

Total no. of Pages:

Roll no.

Degree: B.Tech Semester: III

END-SEMESTER EXAMINATION, NOV-DEC 2025Course Code: EAEPC07
Course Title: Microelectronics

Max. Marks: 40

Duration: 03 Hours

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following	4	1
1a	Explain the concept of Fermi level and its variation with doping.	4	1
1b	Derive the expression for carrier concentration in an intrinsic semiconductor.	4	1
1c	Derive the relationship between drift current and electric field.		
Q 2	Attempt any 2 parts of the following	4	2
2a	Explain the operation of a Zener diode and its use in voltage regulation.	4	2
2b	Determine the output voltage of a half-wave rectifier with $R_L = 1\text{ k}\Omega$ and $V_m = 10\text{ V}$.	4	2
2c	Design a clipper circuit to limit voltage at $\pm 5\text{ V}$.		
Q 3	Attempt any 2 parts of the following	4	3
3a	Explain the concept of thermal runaway and methods to prevent it.	4	3
3b	Determine the small-signal voltage gain of a CE amplifier given parameters.	4	3
3c	Draw and explain the DC load line for a CE amplifier.		
Q 4	Attempt any 2 parts of the following	4	4
4a	Explain the working of enhancement-mode and depletion-mode MOSFETs.	4	4
4b	Discuss short-channel effects in MOSFETs.	4	4
4c	Determine the operating region of a MOSFET for given V_{GS} , V_{DS} , and V_{th} values.		
Q 5	Attempt any 2 parts of the following	4	5
5a	Derive the hybrid- π model of a BJT.	4	5
5b	Find the voltage gain and input/output impedance for a CE amplifier using hybrid- π parameters.	4	5
5c	Describe the high-frequency equivalent circuit of a MOSFET and derive the expression for gain bandwidth product.	4	5

Degree: B.Tech Semester: 7th Sem — Course work
END-SEMESTER EXAMINATION (Odd), Nov-Dec 2025

Course Title: Computer Vision
Course Code: EAEPE53

Max. Marks: 50

Duration: 3 Hours

Note: - Attempt all five questions. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	Explain the objectives of computer vision and give the comparison between Human Visual System and Computer Vision.	5	1
1b	Differentiate between computer vision, image processing and computer graphics	5	1
1c	Briefly describe the applications of computer vision with detailed justification for each.	5	1
Q 2	Attempt any 2 parts of the following		
2a	Explain in detail about image formation and fundamental radiometric quantities.	5	2
2b	Define Gabor filters and how can they be used to extract the texture features of an image.	5	2
2c	Briefly describe histogram processing and image histogram equalisation techniques.	5	2
Q 3	Attempt any 2 parts of the following		
3a	Define point operations, contrast stretching and gray level slicing in detail.	5	3
3b	Describe image enhancement and 2D image representation with suitable examples.	5	3
3c	Explain in brief the types of shape representations in computer vision.	5	3
Q 4	Attempt any 2 parts of the following		
4a	Describe Image pre-processing and Feature Extraction.	5	4
4b	Explain the concept of Principal Component Analysis with diagrams and examples.	5	4
4c	Describe in brief about HMM, GMM and EM.	5	4
	P.T.O.		

Q 5	Attempt any 2 parts of the following		
5a	Explain Face and Facial Expression Recognition in Biometric Applications. How FDLA is efficient in discriminating different face patterns.	5	5
5b	Describe the basic challenges encountered while preparing an algorithm for medical image segmentation.	5	5
5c	List the differences between hierarchical agglomerative clustering and fuzzy C-means clustering.	5	5